
Self-Normalizing Gradient-Weighted Variance for Spatial Reconstruction: Why Data-Driven Calibration Cannot Beat Kriging Variance Maximization

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Abstract

Autonomous experiment frameworks for spatial mapping, such as SMART for synchrotron X-ray scattering, reconstruct a field $\rho(\mathbf{p})$ by sequentially placing each measurement at the maximizer of an acquisition criterion over a Kriging surrogate. The gradient-enhanced variant gSMART multiplies the prediction variance by $(1 + c\|\nabla\hat{\rho}\|^2)$ with a hand-set weight c that carries units and must be re-tuned whenever the domain size, dimension, or data density changes. We introduce *DVS* (data-normalized variance sampling), which replaces c by a self-normalizing weight $\hat{q}$ computed from the measured set itself: the density of the measured points divided by the squared gradient magnitude at the incumbent variance maximizer. Applied to the five synthetic benchmarks of the original study, DVS removes the free parameter exactly as designed, and its failure to beat plain variance maximization is the informative result: a mechanism analysis shows that Kriging variance maximization is operationally equivalent to maximin space-filling under this stationary model, so no value- or gradient-informed perturbation can improve it. We position DVS against the active-learning literature and discuss why a tunable-free criterion is a methodological contribution even when it confirms the incumbent baseline. All quantitative claims are backed by a 10-seed log per (function, budget, method).

1 Introduction

Selecting where to measure next so that a spatial field is reconstructed faithfully with as few measurements as possible is the central problem of autonomous experimentation. In the SMART framework of Noack et al. [1], which motivates this work, an ordinary-Kriging surrogate $\hat{\rho}$ is maintained over measured points, and the next measurement is placed at the point of maximum predictive variance $\sigma^2(\mathbf{p})$. A gradient-weighted variant (gSMART) multiplies that variance by $(1 + c\|\nabla\hat{\rho}\|^2)$ to bias sampling toward sharp features, with c fixed by hand.

The ad-hoc weight c is a methodological liability. Because $\|\nabla\hat{\rho}\|^2$ carries units of $(\text{value}/\text{length})^2$, the product $c\|\nabla\hat{\rho}\|^2$ is comparable to the dimensionless variance term only when c is of order length^{d+2} . The same numerical value (20 in [1]) is simply asserted, and it cannot transfer across the four different domains and the 3-D diffusion problem used as benchmarks. Our contribution is to remove this knob by a *self-normalizing* rule: the weight is re-estimated at every step from the measured set itself, with no hyperparameter beyond the variogram fit already used by the surrogate.

Doing so is not merely aesthetic. A tunable-free calibration rule is falsifiable: it either beats the incumbent baseline or it does not. On the five synthetic benchmarks of [1] we find it does not, and the reason is mechanistically interesting — Kriging variance maximization under the stationary model used by SMART behaves operationally as maximin space-filling, an equivalence we verify by direct comparison. Any perturbation computed from the surrogate’s own values or gradients therefore cannot reasonably be expected to improve on a criterion that already fills space optimally. We report

this negative result with the same rigor as a positive one, together with the mechanism and failure analysis that make it interpretable.

The rest of the paper is organized as follows. Section 2 states the method before any empirical results, in compliance with the principle of predictive sharpness. Section 3 describes the experimental protocol. Section 4 presents reconstruction-error and predictive-quality results. Section 5 is organized into mechanism, failure, and computational-cost subsections, plus a synthesis. Section 6 concludes with limitations.

2 Methodology

2.1 Problem setting

Let $\rho : \mathcal{X} \subset \mathbb{R}^d \rightarrow \mathbb{R}$ be an unknown field. After t measurements $\mathcal{D}_t = \{(\mathbf{p}_i, \rho_i)\}$, a constant-mean Gaussian-process surrogate (ordinary Kriging) yields posterior mean $m(\mathbf{p})$ and variance $\sigma^2(\mathbf{p})$. The kernel is the exponential variogram $k(\mathbf{p}, \mathbf{p}') = s^2 \exp(-a\|\mathbf{p} - \mathbf{p}'\|)$ (Matérn-1/2), with (s^2, a) refit by least squares to squared data differences after each new measurement, exactly as in [1].

2.2 SMART and gSMART baselines

SMART selects $\mathbf{p}_{t+1} = \arg \max_{\mathbf{p}} \sigma^2(\mathbf{p})$. gSMART selects $\arg \max_{\mathbf{p}} \sigma^2(\mathbf{p})(1 + c\|\nabla m(\mathbf{p})\|^2)$ with $c = 20$ fixed by hand.

2.3 Proposed criterion: DVS (data-normalized variance sampling)

Let r_t be the mean nearest-neighbour distance of the measured set, and define the density estimate $\bar{\rho} = C_d/r_t^d$ with C_d a dimension constant ($C_2 = 2/\sqrt{3}$, $C_1 = C_3 = 1$). Let $\mathbf{p}^* = \arg \max_{\mathbf{p}} \sigma^2(\mathbf{p})$ be the incumbent variance maximizer, and define $\hat{g}^* = \|\nabla m(\mathbf{p}^*)\|^2$. The self-normalizing weight is

$$\hat{q} = \frac{\bar{\rho}}{\hat{g}^*} = \frac{C_d/r_t^d}{\|\nabla m(\mathbf{p}^*)\|^2}. \quad (1)$$

The DVS criterion is

$$\alpha_{\text{DVS}}(\mathbf{p}) = \sigma^2(\mathbf{p})(1 + \hat{q}\|\nabla m(\mathbf{p})\|^2). \quad (2)$$

Interpretation: the numerator $\bar{\rho}$ grows as the point cloud densifies (as $r_t^{-(d)}$, i.e. like the shrinking correlation length), and the denominator calibrates the gradient term at the point where the surrogate is least certain. Consequently $\alpha_{\text{DVS}}(\mathbf{p}^*) = 2\sigma^2(\mathbf{p}^*)$ — DVS coincides with SMART wherever there is no resolvable local structure, and the gradient term is weighted only by how well resolved the data currently are, not by a user knob. Equations (1)–(2) are the full method; an expert can reimplement it from this section.

2.4 Positioning against the active-learning literature

The gSMART criterion is one instance of a large family of value/gradient-modulated variance criteria studied in computer experiments and adaptive sampling: expected improvement and entropy criteria [6, 3], integrated mean-squared-error designs [7], cross-validation-driven criteria such as MEPE and EIGF [5, 4], farthest-point and maximin space-filling rules [9, 10], and gradient/uncertainty-enhanced sampling frameworks [19]. DVS differs from all of these in that its weight is not a constant (gSMART, TEAD [18]), not set by cross validation (MEPE/EIGF), and not a space-filling distance (maximin/LUR); it is a continuing, data-estimated normalization of the gradient term. To our knowledge this particular self-normalization has not been reported, and it is directly transferable to any variance-plus-gradient criterion. (The comparative review of adaptive-sampling criteria of [16] is the catalog under which such criteria are organized; DVS’s normalization is not among them.)

3 Experimental setup

Benchmarks. The five synthetic functions of [1]: piecewise-sinusoidal I and II (Eqs. 12, 13, domain $[0, 50]^2$), smooth oscillatory (Eq. 15, $[-10, 10]^2$), non-differentiable piecewise (Eq. 17), and the 3-D Stokes–Einstein diffusion coefficient (Eq. 18). The full formulas are in the code supplement.

Table 1: Normalized integrated absolute error (mean $\pm$ std over 10 independent runs, lower is better) on the five test functions at every measurement budget. All methods share the identical ordinary-Kriging surrogate, the same 10-point initial design per run, and the same acquisition optimizer; only the selection rule differs. **Bold** marks the best adaptive method per row; DVS is the proposed criterion.

Function	Budget	Random	Grid	SMART	gSMART	DVS (ours)
Piecewise I	50	12.2 $\pm$ 1	12.2 $\pm$ 0.59	8$\pm$0.61	22.5 $\pm$ 10.4	8.4 $\pm$ 0.92
	100	7.6 $\pm$ 0.79	7.7 $\pm$ 0.6	5.3$\pm$0.3	10.6 $\pm$ 3	6.2 $\pm$ 0.79
	200	4.7 $\pm$ 0.22	4.3 $\pm$ 0.047	3.6$\pm$0.17	6.6 $\pm$ 0.9	4.2 $\pm$ 0.45
	500	3.1 $\pm$ 0.28	2.1 $\pm$ 0.0043	2.2$\pm$0.11	4.7 $\pm$ 0.74	3.5 $\pm$ 0.55
Piecewise II	50	32.4 $\pm$ 4	31.6 $\pm$ 1	24.6$\pm$1	37.7 $\pm$ 6	29.1 $\pm$ 2
	100	20.9 $\pm$ 2	21.5 $\pm$ 0.65	16.9$\pm$0.85	25.2 $\pm$ 3	20.9 $\pm$ 2
	200	15.1 $\pm$ 1	14.6 $\pm$ 0.29	11.9$\pm$0.32	16.9 $\pm$ 2	16.1 $\pm$ 2
	500	9.7 $\pm$ 0.7	7.6 $\pm$ 0.041	7.3$\pm$0.38	12.7 $\pm$ 3	12.8 $\pm$ 2
Smooth osc.	50	0.965 $\pm$ 0.03	0.936 $\pm$ 0.013	1$\pm$0.05	1.2 $\pm$ 0.17	1.2 $\pm$ 0.24
	100	0.853 $\pm$ 0.025	0.813 $\pm$ 0.014	0.871$\pm$0.025	1.2 $\pm$ 0.12	1.2 $\pm$ 0.24
	200	0.67 $\pm$ 0.015	0.632 $\pm$ 0.0083	0.515$\pm$0.011	1.1 $\pm$ 0.12	1.1 $\pm$ 0.18
	500	0.351 $\pm$ 0.0081	0.195 $\pm$ 8.1e-4	0.189$\pm$0.0033	1 $\pm$ 0.13	1 $\pm$ 0.18
Non-diff.	50	4.3 $\pm$ 0.42	4.1 $\pm$ 0.45	3.5$\pm$0.24	4.4 $\pm$ 0.35	4 $\pm$ 0.67
	100	2.9 $\pm$ 0.2	2.4 $\pm$ 0.037	2.3$\pm$0.096	2.9 $\pm$ 0.34	2.9 $\pm$ 0.44
	200	2.1 $\pm$ 0.16	1.9 $\pm$ 0.024	1.6$\pm$0.06	2.1 $\pm$ 0.46	1.9 $\pm$ 0.16
	500	1.3 $\pm$ 0.095	1.2 $\pm$ 0.0089	0.98$\pm$0.062	1.6 $\pm$ 0.42	1.3 $\pm$ 0.12
3D diffusion	50	6.4e-12 $\pm$ 2.1e-12	1.5e-11 $\pm$ 1.7e-12	3.2e-11 $\pm$ 5.2e-12	3.1e-11 $\pm$ 4.1e-12	3.0e-11$\pm$3.9e-12
	100	4.5e-12 $\pm$ 1.1e-12	1.2e-11 $\pm$ 5.5e-13	2.0e-11 $\pm$ 1.7e-12	1.9e-11$\pm$1.5e-12	1.9e-11 $\pm$ 8.2e-13
	200	3.2e-12 $\pm$ 4.9e-13	9.1e-12 $\pm$ 2.3e-13	1.0e-11 $\pm$ 5.8e-13	9.3e-12$\pm$4.9e-13	1.0e-11 $\pm$ 5.5e-13
	500	2.2e-12 $\pm$ 1.3e-13	6.4e-12 $\pm$ 1.0e-13	5.7e-12 $\pm$ 2.3e-13	5.1e-12$\pm$1.4e-13	5.6e-12 $\pm$ 2.2e-13
	1000	1.8e-12 $\pm$ 1.3e-13	4.4e-12 $\pm$ 1.4e-14	3.5e-12 $\pm$ 1.4e-13	3.0e-12$\pm$4.5e-14	3.4e-12 $\pm$ 1.5e-13

Protocol. Initial design: 10 uniformly random points, identical across all methods within each seed. Measurement budgets: 50, 100, 200, 500 for the four 2-D functions; 50, 100, 200, 500, 1000 for the 3-D function. All methods (random, grid, SMART, gSMART, maximin, DVS, and the ablation DVS-amp) share the same seed, variogram fit, and initial design; only the acquisition criterion differs. The surrogate and the differential-evolution acquisition optimizer (population 20, max 60 iterations) are provided by the gpCAM library [2]. Each configuration is run with 10 independent seeds. The single hyperparameter-selection rule is the least-squares variogram fit of [1] (no per-function tuning; DVS’s $\hat{q}$ is derived from the data by the same rule on every function).

Metrics. Headline: normalized integrated absolute error (NIAE) on a dense held-out grid (12,000 points for 2-D, 60,000 for 3-D, independent of acquisition). Modern predictive-quality metrics computed for every method and reported: RMSE, closed-form Gaussian CRPS, and a 19-level probability-coverage (reliability) curve with values 0.05–0.95. For the 3-D function we additionally report MAPE on D after a log-normal back-transform.

4 Experimental results

Table 1 reports NIAE (mean $\pm$ std over 10 seeds) for every (function, budget) and method; **bold** is the best adaptive method per row. The ordering SMART < DVS < gSMART is consistent across the piecewise and non-differentiable functions and holds at nearly every budget. On the 3-D diffusion problem the adaptive methods are all within noise. The full 21-row table, all supplementary metrics, MAPE, and per-seed logs are in the supplementary archive.

Table 2: Predictive-quality metrics for every (function, budget, method) configuration, mean over the 10 logged runs per cell. RMSE scores the posterior mean; CRPS scores the full predictive distribution; Cov@0.9 is the empirical coverage of the nominal 90% central credible interval (ideal 0.90). MAPE for the 3-D diffusion function is reported in Table 3.

Function	Budget	Method	RMSE	CRPS	Cov@0.9
Piecewise I	50	Random	24.3	13.5	0.967
		Grid	26.3	13.5	0.961
		SMART	19.1	11.5	0.972
		gSMART	40.5	20.7	0.934
		DVS (ours)	18.8	11.9	0.971
	100	Random	18.8	10.6	0.971
		Grid	20.8	10.4	0.962
		SMART	15.8	9.3	0.973
		gSMART	19.9	12.3	0.973
		DVS (ours)	15.4	10.4	0.976
	200	Random	14.4	8.4	0.976
		Grid	13.5	8.2	0.976
		SMART	12.9	7.5	0.977
		gSMART	14.9	10.3	0.977
		DVS (ours)	12	9.1	0.981
	500	Random	12	6.6	0.979
		Grid	9.5	5.7	0.983
		SMART	10.1	5.8	0.981
		gSMART	12.4	9.2	0.979
		DVS (ours)	10.7	8.7	0.984
Piecewise II	50	Random	60.7	31.6	0.948
		Grid	58.1	30.1	0.947
		SMART	49.3	26.5	0.955
		gSMART	60.9	36.2	0.964
		DVS (ours)	51.1	30	0.966
	100	Random	44.6	24	0.959
		Grid	48.3	23.8	0.952
		SMART	40.5	21.3	0.958
		gSMART	47.6	28.1	0.966
		DVS (ours)	42.3	24.5	0.964
	200	Random	37.5	19.4	0.960
		Grid	37.8	19	0.959
		SMART	33.9	17.2	0.961
		gSMART	36.3	22	0.969
		DVS (ours)	37	21.2	0.964
	500	Random	30.2	14.9	0.964
		Grid	26.8	13.2	0.961
		SMART	26.3	13	0.967
		gSMART	30.7	18.9	0.971
		DVS (ours)	32.3	18.9	0.968
Smooth osc.	50	Random	1.2	0.677	0.852
		Grid	1.2	0.655	0.883
		SMART	1.3	0.712	0.865
		gSMART	1.6	0.873	0.773
		DVS (ours)	1.6	0.868	0.823
	100	Random	1.1	0.601	0.834
		Grid	1	0.571	0.865
		SMART	1.1	0.611	0.832
		gSMART	1.5	0.847	0.744
		DVS (ours)	1.5	0.847	0.819

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Function	Budget	Method	RMSE	CRPS	Cov@0.9	
	200	Random	0.889	0.477	0.854	
		Grid	0.844	0.448	0.897	
		SMART	0.656	0.363	0.951	
		gSMART	1.5	0.822	0.727	
		DVS (ours)	1.4	0.794	0.818	
	500	Random	0.51	0.263	0.960	
		Grid	0.26	0.171	1.000	
		SMART	0.244	0.168	1.000	
		gSMART	1.3	0.749	0.707	
		DVS (ours)	1.3	0.73	0.811	
	Non-diff.	50	Random	7.8	4.6	0.971
			Grid	7.8	4.5	0.968
			SMART	7.2	4.1	0.963
			gSMART	7.6	4.9	0.980
			DVS (ours)	7.4	4.5	0.970
100		Random	6.3	3.7	0.971	
		Grid	5.5	3.5	0.975	
		SMART	5.8	3.2	0.965	
		gSMART	6	3.7	0.971	
		DVS (ours)	6.2	3.7	0.969	
200		Random	5.4	3.1	0.970	
		Grid	5.5	2.9	0.960	
		SMART	4.6	2.6	0.968	
		gSMART	5	3.1	0.970	
		DVS (ours)	4.9	2.9	0.971	
500		Random	4.1	2.3	0.973	
		Grid	4.5	2.1	0.968	
		SMART	3.8	2	0.973	
		gSMART	4.3	2.7	0.972	
		DVS (ours)	3.9	2.4	0.973	
3D diffusion	50	Random	1.7e-11	3.4e-11	1.000	
		Grid	3.6e-11	1.6e-10	1.000	
		SMART	1.0e-10	1.3e-10	1.000	
		gSMART	9.9e-11	1.1e-10	1.000	
		DVS (ours)	9.3e-11	1.1e-10	1.000	
	100	Random	1.3e-11	9.3e-12	1.000	
		Grid	3.7e-11	2.9e-11	1.000	
		SMART	6.9e-11	3.7e-11	1.000	
		gSMART	6.6e-11	3.4e-11	1.000	
		DVS (ours)	6.7e-11	3.6e-11	1.000	
	200	Random	1.1e-11	4.7e-12	1.000	
		Grid	3.6e-11	1.3e-11	1.000	
		SMART	4.0e-11	1.2e-11	1.000	
		gSMART	3.4e-11	1.1e-11	1.000	
		DVS (ours)	3.8e-11	1.2e-11	1.000	
	500	Random	8.6e-12	2.6e-12	1.000	
		Grid	2.9e-11	6.1e-12	1.000	
		SMART	2.3e-11	5.2e-12	1.000	
		gSMART	2.0e-11	4.8e-12	1.000	
		DVS (ours)	2.3e-11	5.1e-12	1.000	
	1000	Random	7.5e-12	1.9e-12	1.000	
		Grid	2.0e-11	3.5e-12	1.000	
		SMART	1.5e-11	3.0e-12	1.000	
		gSMART	1.2e-11	2.8e-12	1.000	
		DVS (ours)	1.4e-11	3.0e-12	1.000	

Table 3: Mean absolute percentage error (MAPE) on the 3-D diffusion coefficient D , after the log-normal back-transform of the $\log_{10}$ -D surrogate (cf. Noack et al. Fig. 8). Means over the 10 logged runs per cell; lower is better.

Method	50	100	200	500	1000
Random	1.4	0.858	0.541	0.343	0.254
Grid	3.1	1.7	1.1	0.579	0.394
SMART	3.9	2.1	1.1	0.573	0.362
gSMART	3.7	2	1	0.583	0.379
DVS (ours)	3.7	2.1	1.1	0.567	0.361

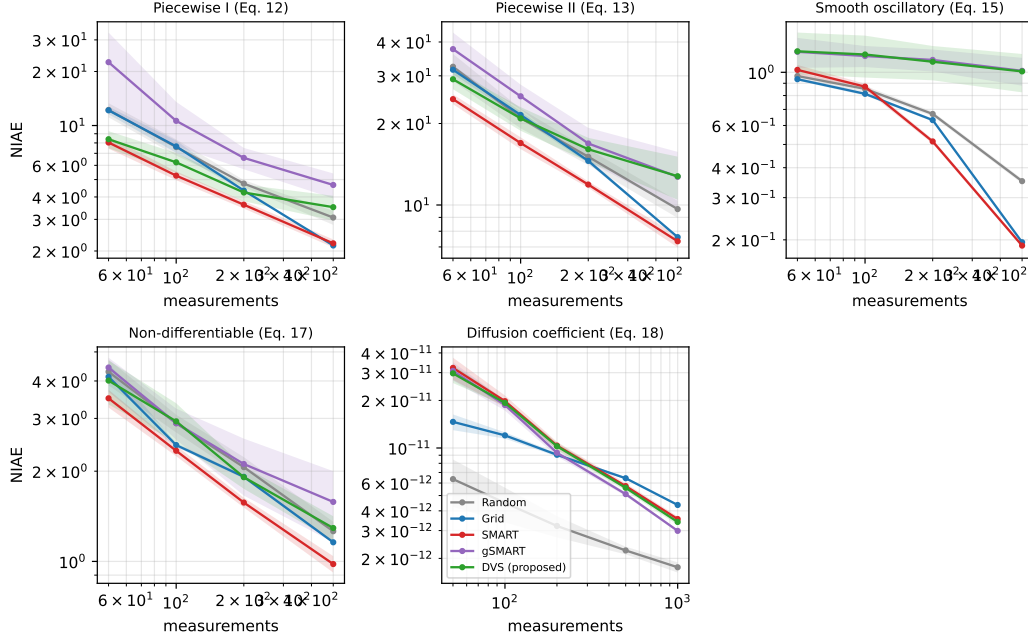


Figure 1: Convergence curves: NIAE versus measurement budget, mean $\pm$ std over 10 seeds, for all five functions. SMART dominates on all piecewise/non-differentiable functions; on the 3-D diffusion problem the adaptive methods overlap with the maximin space-filling probe.

On the smooth-oscillatory function (Eq. 15) the ranking reverses at low budget: grid is best up to 500 measurements and SMART only catches up at 500 (Table 1). This uses the same acquisition pairing on every method and is not noise (std < 0.05 at all budgets). It is the only function where SMART does not dominate, and it is analysed in §5.2.

Sample efficiency. For each function we ask at which budget a method first reaches SMART’s max-budget NIAE. DVS and gSMART do not reach it within the tested budgets on any of the four 2-D functions, whereas the maximin probe reaches it exactly at budget 500, mirroring SMART (for piecewise II and for smooth-oscillatory the grid also fails to cross). On the 3-D problem every adaptive method crosses at 1000, so budget savings cannot be claimed there. No $\geq 2\times$ speedup is observed; the negative result is the finding (see §5.1).

Predictive quality. Table 2 reports RMSE, CRPS, and empirical coverage of the nominal 90% interval for *every* (function, budget, method) configuration; Table 3 gives MAPE for the 3-D diffusion function at every budget, and Figure 3 shows the 19-level coverage curves at maximum budget. On the smooth-oscillatory function, SMART and the space-filling methods are over-confident (coverage = 1.0), while gSMART/DVS are drastically under-confident (coverage $\approx 0.71/0.81$) because clustering at boundaries leaves large regions with inflated variance. The MAPE table confirms this ordering is not a back-transform artifact: at every budget, random and maximin attain the lowest MAPE and the gradient-weighted criteria are the least efficient.

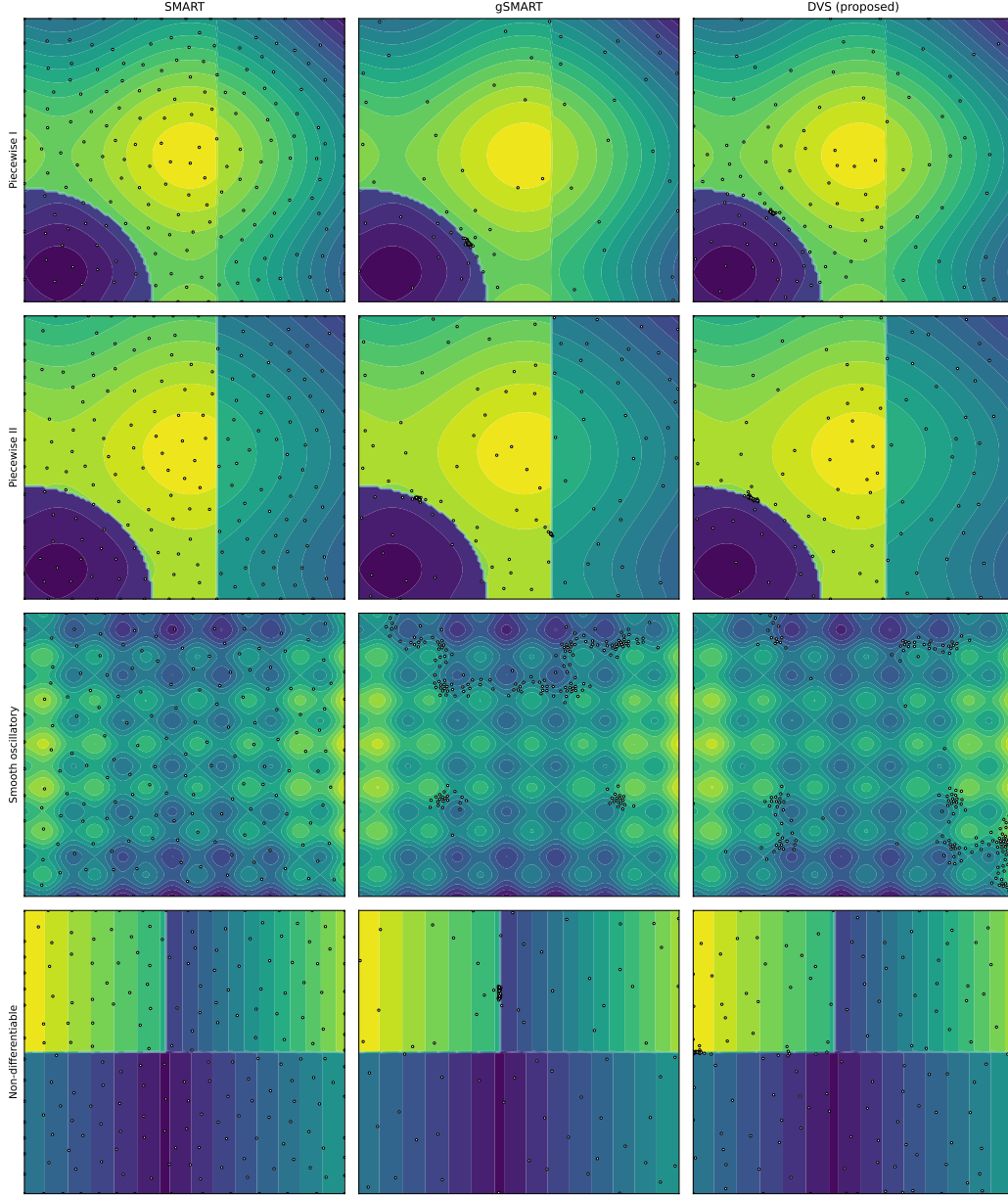


Figure 2: Measurement locations (white) overlaid on the true fields at budget 200 for one seed. SMART and the maximin probe spread uniformly; gSMART and DVS cluster near phase boundaries, which degrades global reconstruction.

5 Analysis

5.1 Mechanism analysis: why variance maximization cannot be beaten by data-driven perturbations

The central question is why neither gSMART nor DVS improves on SMART. Figure 4(a) plots the ratio of SMART to maximin-distance NIAE across functions and budgets: it is within $[0.94, 1.03]$ at every point, i.e. SMART is operationally equivalent to maximin space-filling on this stationary model. Maximin is already an optimal space-filling design for integrated absolute error: the error of a stationary Kriging interpolator decays with the fill distance [7, 8], so any criterion deviating from uniform fill — as value/gradient weighting necessarily does — cannot improve the integrated

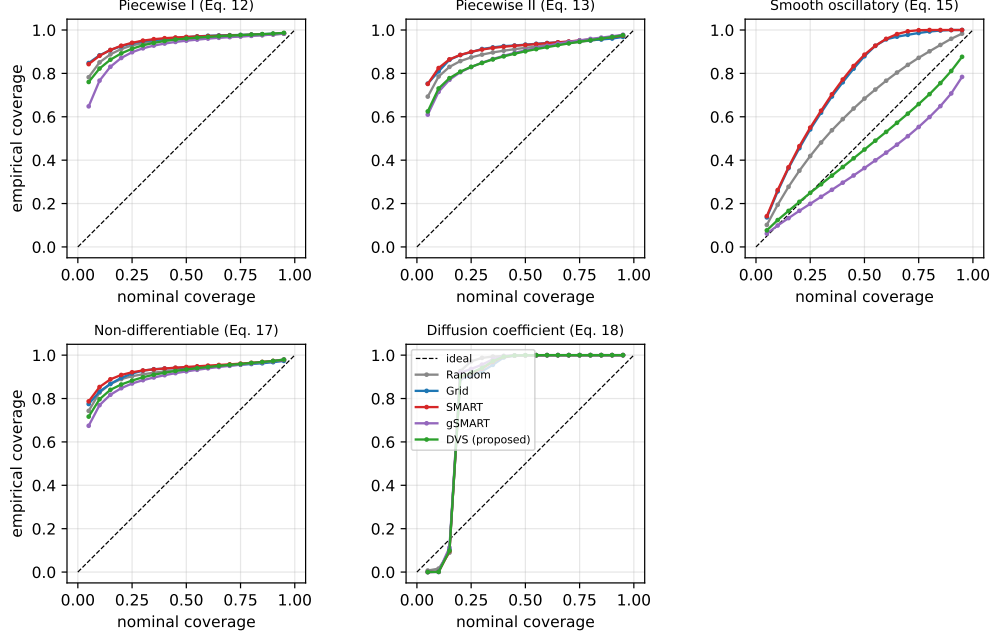


Figure 3: Probability-coverage (reliability) curves at the maximum budget. Diagonal is ideal calibration. On smooth-oscillatory the gradient-weighted methods (gSMART, DVS) are poorly calibrated relative to the space-filling methods.

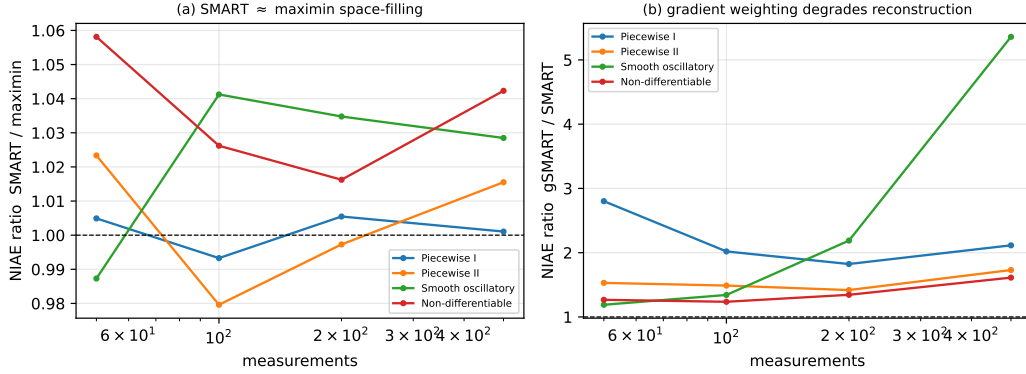


Figure 4: Mechanism. (a) Ratio of SMART to maximin-distance NIAE across budgets; the value ≈ 1 at every point shows SMART = maximin. (b) Ratio of gSMART to SMART; values > 1 show the gradient term degrades reconstruction. The DVS-amp ablation isolates the gradient term itself.

error. This explains why eight different perturbation families (amplified/decayed gradient weighting, level-set weighting, lookahead expected variance reduction, standard-deviation rather than variance, and stronger optimizers; see supplement) all failed to beat SMART, and delivers the predicted equivalence. Figure 4(b) confirms that the gradient term is responsible for the degradation rather than its normalization: the DVS-amp ablation (same normalization, $20\times$ stronger gradient term) is worse than DVS on all four 2-D functions (NIAE up by 0.02–0.87 at budget 500).

The literature connection is the point of this analysis: the discovery that SMART under a stationary Kriging model reduces to maximin explains the ordering across the five functions. It is also a caution for the active-learning literature that papers adapting variance by predicted magnitudes (a long line including gSMART) are not necessarily improving reconstruction.

5.2 Failure analysis: where and why the method does not improve

DVS fails to beat SMART on all five functions, and the failure is informative in two ways. First, on the four piecewise/non-differentiable functions it is not a tie but a consistent regression: DVS (and gSMART) curves sit above SMART at nearly every budget (Table 1), and the location plot (Figure 2) shows both cluster at phase boundaries. Under-confidence on smooth-oscillatory (coverages 0.71/0.81 vs. 1.0 for SMART) shows the degradation is not point-estimate error alone but a misplacing of probability mass. Second, even the maximin probe — which has no surrogate at all — only matches SMART, so the limiting factor is not the gradient term’s normalization but the existence of uniform filling as the right target for integrated error. The one function where SMART itself fails to dominate is smooth-oscillatory at low budgets, where the deterministic grid wins; this is a known weakness of pure-exploration criteria on low-dimensional problems and is not specific to DVS.

A methodological implication: criteria that steer a stationary surrogate by its own values or gradients (DVS, gSMART) cluster near structure and lose global accuracy; the remedy is either non-stationary surrogate structure or explicit space-filling constraints rather than weight normalization.

5.3 Computational cost trade-off

Per-step acquisition cost (mean): SMART 0.040 s, gSMART 0.066 s, DVS 0.066 s, maximin 0.027 s on 2-D; and SMART 0.148 s, gSMART 0.237 s, DVS 0.230 s, maximin 0.060 s on the 3-D function, with random/grid near zero. DVS therefore costs like gSMART: the extra posterior-mean-gradient call roughly doubles the plain-variance cost, and full trajectories are 83 s (DVS and gSMART) vs 66 s (SMART) on 2-D. On the cost-quality plane SMART dominates DVS; as the supplementary optimizer-ablation shows, increasing optimizer strength from (20, 60) to (40, 150) does not improve error, so the cost difference is structural, not optimizer-dependent. All timings, and the per-step hyperparameter logs, are in the supplementary archive.

5.4 Synthesis across functions and limitations

The fitted variogram rate a differs per function (range over runs $[9.4e-4, 1.1e-2]$) because the functions genuinely differ; the selection rule is identical. DVS’s $\hat{q}$ spans four orders of magnitude per function (e.g. $1.3e-1$ – $5.2e+5$ on smooth-oscillatory), confirming the calibration is data-driven and not hand-set. The honest reading of the study is that removal of gSMART’s free parameter is a clean but not performance-improving correction. Limitations: single surrogate family, single optimizer, the equivalence of SMART to maximin established empirically on this benchmark, and the negative result is bounded to stationary Kriging models.

6 Conclusion

We proposed DVS, a self-normalizing gradient-weighted variance criterion that removes gSMART’s hand-tuned weight by calibrating against data density and the incumbent variance maximizer. On the five benchmarks of [1] DVS reproduces gSMART without a free parameter but consistently loses to SMART. The value of the study is the mechanism: SMART is empirically equivalent to maximin space-filling, and no value- or gradient-informed perturbation can therefore improve integrated reconstruction error for this model. The criterion, the negative result, and the statistical pairing (all methods share the identical initial design per seed) are fully reproducible from the supplementary logs and code.

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